

Oluwatobi Melvyn Mayungbo

Data Analyst, Fraud and Risk

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SUMMARY

Fraud analyst with a B.Sc. in Statistics and internal audit experience. Built fraud detection that breaks results down by attack type, an explainable risk scorer with a reason for every point, and a reconciliation system that flags duplicate settlements and payout discrepancies. Reviewed financial records for irregularities as an internal auditor.

SKILLS

Payments and Risk: Transaction Fraud Detection, Payment Reconciliation, Regulatory Compliance (CBN/NDPA), Credit Risk Decisioning, Double-Entry Ledgers and Idempotency, Demand Forecasting, Web3 Fraud Risk Scoring

Data Analysis: SQL, Pandas, Anomaly Detection, NumPy

Machine Learning: LightGBM, Feature Engineering, PyTorch, Scikit-learn, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP

Engineering and MLOps: Python, Streamlit, SQL, DuckDB, PostgreSQL, Redis, FastAPI, Docker, GCP Cloud Run, AWS ECS Fargate, Railway, Prefect, dbt-core, MLflow, Evidently, CI/CD (GitHub Actions), Git, Pytest, React

Statistics and Causal Inference: Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

AI and LLM Systems: Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), LLM Evaluation and Guardrails, Prompt Engineering, Citation Verification, Model Context Protocol (MCP), Hugging Face, ONNX Runtime

PROJECTS

Sequence-Based Transaction Fraud Detection System | [GitHub](#)

Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP

- Benchmarked all architectures against a strong LightGBM baseline on engineered velocity and deviation features, reporting PR-AUC and precision at fixed review capacity (top 0.1%, 0.5%, 1%) across 3 seeds, with a per-attack-type breakdown covering card testing, account takeover and merchant compromise; sequence models cut expected cost per transaction 27% under an explicit cost model.
- Built and compared three deep sequence architectures (GRU, Temporal Convolutional Network and a causally masked Transformer encoder) over customer transaction histories, with learned embeddings for high-cardinality merchant (~2.9k vocabulary), MCC and device-type categoricals; all models under 115k parameters and CPU-trainable in under 6 minutes.

Reckon: Payment Reconciliation and Review System | [GitHub](#) | [Live App](#)

Tech: Python, PyTorch, DuckDB, FastAPI, React, Docker, Railway, PostgreSQL

- Built and deployed a payment reconciliation system processing multi-source transaction feeds to detect uncollected revenue, duplicate settlements and payout discrepancies, combining deterministic matching rules with ML-assisted discrepancy resolution.
- Designed a human-in-the-loop review interface in React showing match confidence scores, audit trails and suggested resolution actions to streamline exception handling.

web3-risk-mcp: Web3 Fraud Risk MCP Server | [GitHub](#)

Tech: Python, Model Context Protocol (MCP), Etherscan V2, GoPlus, DexScreener, Docker, GitHub Actions

- Designed a rule-based, fully explainable scorer where findings carry stable IDs and decisive findings set a floor of 75 so trust signals can never hide them; each score carries a confidence level based on how many sources answered.
- Detects honeypot tokens, rug pull signs, hidden owner powers (mint, blacklist, pause), buy and sell tax, holder concentration, unlocked liquidity, and fund links to mixers, sanctioned wallets, exploiters and phishing addresses over 1 or 2 hops.

EXPERIENCE

Internal Auditor, NISER

May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

EDUCATION

B.Sc. Statistics, University of Ibadan

January 2021 to March 2025

Professional Diploma in Artificial Intelligence, SQI College of ICT

July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action